

# MERIDIAN & ASSOCIATES LLP

Audit & Assurance Practice

## Audit Technology Integration Framework

Proprietary AI, Analytics, and Automation Tools  
for Financial Statement Audits

**Document ID:** MA-AA-TI-2026-006

**Version:** 3.0

**Effective Date:** January 30, 2026

**Classification:** CONFIDENTIAL

**Approved By:** National Assurance Quality Board

**Next Review:** January 2027

### CONFIDENTIAL

This document contains proprietary methodologies and procedures of Meridian & Associates LLP. It is intended solely for use by authorized personnel. Distribution outside the firm requires written approval from the National Managing Partner, Assurance Services.

## 1. Introduction and Strategic Vision

---

Meridian & Associates LLP has made a strategic commitment to integrating advanced technology into every phase of the audit process. The firm has invested over \$240 million in its audit technology platform since 2020, building a suite of proprietary tools designed to enhance audit quality, improve efficiency, and deliver deeper insights to audit clients and their stakeholders. This framework document describes the firm's current technology capabilities and their integration into the standard audit methodology.

The firm's technology strategy is guided by three core principles: (1) technology should augment professional judgment, not replace it; (2) full-population testing should be the default where technically feasible, replacing traditional sampling; and (3) all technology tools must be validated, documented, and subject to the same quality standards as traditional audit procedures.

The firm's technology platform comprises two complementary pillars: MeridianAudit Suite, the engagement documentation and workflow platform used to manage work papers, review notes, and sign-offs; and MeridianAI, the analytics and artificial intelligence platform (including MeridianAI Analyze and MeridianAI Contract Reader) used for data-driven audit procedures such as journal entry testing, fraud risk scoring, and contract analysis.

## 2. MeridianAI Analyze -- Journal Entry Testing

---

### 2.1 Capabilities

MeridianAI Analyze is the firm's flagship AI-powered audit tool for journal entry testing. Unlike traditional sampling-based approaches (which typically test fewer than 1% of journal entries), MeridianAI Analyze tests 100% of journal entries recorded during the audit period. The platform processes entries from any major ERP system (SAP, Oracle, Workday, NetSuite, Microsoft Dynamics) and uses a multi-layered analysis approach:

- Rule-Based Screening: Identifies entries matching predefined risk indicators, including entries posted after business hours, entries by users with unusual posting authority, entries with round-dollar amounts exceeding specified thresholds, entries posted with non-standard or blank descriptions, and entries posted at or near period-end with reversals in the subsequent period.
- Machine Learning Anomaly Detection: A supervised learning model trained on the firm's proprietary dataset of over 2.8 billion historical journal entries identifies entries that deviate from expected patterns based on the entity's historical posting behavior. The model considers account combinations, amounts, timing, frequency, and user identity.
- Natural Language Processing (NLP): Analyzes the text descriptions of journal entries to identify entries with unusual or inconsistent narratives, entries referencing related parties or key management personnel, and entries with descriptions that are inconsistent with the affected accounts.
- Network Analysis: Maps relationships between posting users, accounts, and amounts to identify complex entry chains that may indicate attempts to obscure the true nature of transactions.

### 2.2 Validation and Audit Evidence

The firm's Data Science Quality Board has validated MeridianAI Analyze's models using back-testing against known fraud cases and restatements. The tool's precision rate (correct identification of unusual entries) exceeds 92%, while the recall rate (identification of all genuinely unusual entries) exceeds 98%. All model outputs are reviewed by qualified audit professionals before being used as audit evidence. The tool does not make audit judgments; it identifies entries for further investigation.

## 3. Predictive Analytics for Fraud Risk Scoring

---

The firm has developed a proprietary fraud risk scoring model that supplements the engagement team's fraud risk assessment under PCAOB AS 2401 and ISA 240. The model evaluates entity-level and account-level fraud risk factors using a combination of:

- Financial statement analytics: Beneish M-Score, Altman Z-Score, revenue quality indicators, earnings management proxies, and accrual-based measures.
- Non-financial indicators: Management turnover, auditor changes, related-party transaction complexity, governance quality scores, litigation and regulatory history, and media sentiment analysis.

- Industry benchmarking: Comparison of key financial metrics against industry peer groups using the firm's proprietary database of over 15,000 public company financial profiles.
- Temporal analysis: Trend analysis across multiple reporting periods to identify deteriorating patterns that may indicate elevated fraud risk.

The model produces a composite fraud risk score (1-100 scale) for each engagement, with scores categorized as Low (1-30), Moderate (31-60), High (61-80), and Very High (81-100). Engagements scoring above 60 require mandatory consultation with the firm's Forensic and Litigation Services practice.

## 4. Automated Three-Way Matching

---

The firm's automated three-way matching tool digitally matches purchase orders, receiving reports, and vendor invoices for accounts payable testing. The tool operates on the client's procurement data (extracted via secure API connections to the client's ERP system) and performs the following procedures without manual intervention:

- Matching of purchase order, goods receipt, and invoice across quantity, price, and amount fields with configurable tolerance thresholds.
- Identification of unmatched or partially matched transactions for targeted follow-up.
- Analysis of duplicate payments, duplicate vendor records, and payments to vendors not on the approved vendor master file.
- Evaluation of purchase order authorization levels and identification of split orders that may indicate circumvention of approval thresholds.
- Aging analysis of unmatched receiving reports to identify potential unrecorded liabilities.

## 5. Drone and Satellite Imagery for Inventory Observation

---

For engagements involving physically large or geographically dispersed inventory (e.g., mining, agriculture, energy, real estate), the firm supplements traditional physical observation procedures with drone and satellite imagery:

- Drone-mounted LiDAR (Light Detection and Ranging) sensors create three-dimensional volumetric measurements of bulk inventories such as stockpiles, grain silos, and liquid storage tanks. These measurements are compared to management's recorded quantities with a measurement accuracy of +/- 1.5%.
- Commercial satellite imagery (sub-meter resolution) is used to corroborate the existence and scale of large-area inventory, including timber stands, agricultural crops, and solar panel installations. Multi-temporal analysis compares imagery from different dates to assess changes in inventory levels.
- All drone and satellite procedures are performed under the supervision of qualified audit professionals. The imagery and volumetric data constitute corroborative audit evidence and are documented in the firm's standard work paper format.

## 6. NLP for Contract Analysis

---

The firm's NLP-powered contract analysis tool, MeridianAI Contract Reader, automates the review of large volumes of contracts for key audit-relevant terms. The tool is used extensively in the following audit areas:

- Revenue recognition (ASC 606): Extraction of performance obligations, variable consideration terms, significant financing components, and contract modification provisions.
- Lease accounting (ASC 842): Identification of lease terms, renewal and termination options, variable lease payments, residual value guarantees, and embedded lease arrangements.
- Debt and financing (ASC 470/ASC 815): Extraction of covenant terms, interest rate provisions, prepayment penalties, conversion features, and embedded derivatives.
- The tool processes contracts in PDF, Word, and scanned image formats (via OCR). It has been trained on over 500,000 contracts across 22 industry sectors and achieves a field-level extraction accuracy exceeding 95%.

## 7. RPA for Confirmations Processing

---

The firm has deployed Robotic Process Automation (RPA) to streamline the audit confirmations process. The RPA solution handles:

- Automated generation and distribution of confirmation requests (bank confirmations, accounts receivable confirmations, accounts payable confirmations, and legal confirmations) using data extracted from client systems.
- Automated tracking of outstanding confirmations with escalation workflows at 10, 20, and 30 days past the request date.
- Digital receipt and reconciliation of returned confirmations, with automatic matching of confirmed balances to the general ledger. Discrepancies exceeding the de minimis threshold are flagged for manual investigation.
- Integration with the AICPA Confirmation.com platform for electronic bank and other financial institution confirmations.
- Year-over-year response rate analytics to identify clients or third parties with historically low response rates, enabling the team to plan alternative procedures proactively.

## 8. Data Lake Architecture and ERP Integration

---

The firm's Audit Data Lake provides a centralized, secure repository for all client data extracted during the audit process. Key features include:

- Pre-built connectors for major ERP systems: SAP (S/4HANA and ECC), Oracle (Cloud ERP and E-Business Suite), Workday, NetSuite, Microsoft Dynamics 365, and Infor.
- Automated data extraction, transformation, and loading (ETL) pipelines that standardize client data into the firm's common audit data model, enabling consistent analytics across engagements.
- Role-based access controls ensuring that engagement team members only access data for their assigned engagements, with full audit trails of all data access and modifications.
- Data retention policies compliant with PCAOB AS 1215 (Audit Documentation) and SEC Rule 2-06 of Regulation S-X (seven-year retention) and applicable data privacy regulations.
- Encryption at rest (AES-256) and in transit (TLS 1.3), with data residency options to comply with cross-border data transfer requirements.